

Eclipse Kura works on the Java Virtual Machine (JVM) and utilises OSGi, a dynamic component system for Java, to streamline the development of reusable software building blocks. The Kura APIs provide access to underlying hardware components such as serial ports, GPS, watchdog timers, USB, GPIOs, I2C, and more. Additionally, Kura offers OSGi bundles that simplify network configuration management, communication with IoT servers, and remote gateway administration.

Kura components are implemented as configurable OSGi Declarative Services, which expose service APIs and generate events. While many Kura components are written purely in Java, some rely on JNI (Java Native Interface) and require a Linux operating system to function. Kura comes with the services shown in Figure 1.

Before installing Kura, ensure the following requirements are met.

**Hardware requirements:** A compatible gateway or device (e.g., Raspberry Pi, industrial IoT gateway) with sufficient storage and memory to run the Java Virtual Machine (JVM).

**Operating system:** A supported Linux-based operating system (e.g., Ubuntu, Raspbian, or Debian). Ensure that the device is running a 32-bit or 64-bit Linux OS as required.

**Java Virtual Machine (JVM):** Java Development Kit (JDK) 11 or later. Ensure that Java is installed and properly configured. Verify the installation using the `java -version` command.

**Network configuration:** Internet access for downloading Kura packages and dependencies.

**Tools and dependencies:** Tools like SSH for remote access (optional).

**MQTT broker:** Install or use an MQTT broker like Eclipse Mosquitto for testing.

## Setting up Eclipse Kura

Install Eclipse Kura on your target IoT gateway or test on an emulator like Raspberry Pi. Access the Kura Web UI by entering the gateway's IP address and the default port in your browser (e.g., `http://<ip_address>:8080`). Next, log in with the default credentials as mentioned below:

Username: admin  
Password: admin

## Creating a simple application

Install the Eclipse IDE for Java developers. Add the Eclipse Kura development environment using the Kura plugins.

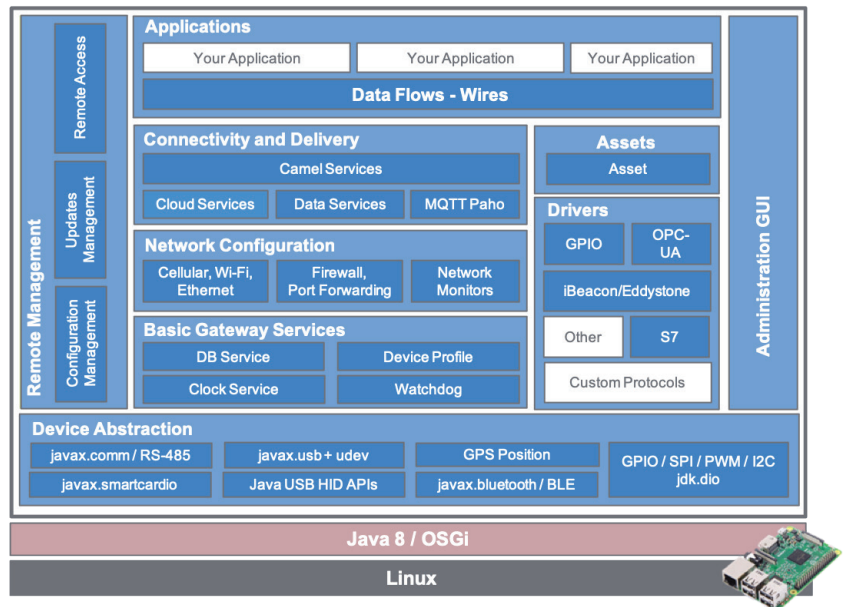


Figure 1: Kura services

To create a new Maven project, open Eclipse. Go to `File > New > Maven Project`. Choose a suitable group ID and artifact ID (e.g., `com.example.osfy`).

Add Kura dependencies by updating the `pom.xml` file with the following:

```
dependencies.xml
<dependencies>
  <dependency>
    <groupId>org.eclipse.kura</groupId>
    <artifactId>org.eclipse.kura.api</artifactId>
    <version>5.1.0</version> <!-- Replace with your
version -->
  </dependency>
</dependencies>
```

To develop a simple Kura application in Java code, create a class `SensorPublisher` that implements `org.eclipse.kura.application.KuraApplication`.

```
package com.example.kuraapp;
import org.eclipse.kura.KuraException;
import org.eclipse.kura.application.KuraApplication;
import org.eclipse.kura.configuration.ConfigurableComponent;
import org.eclipse.kura.message.KuraPayload;
import org.eclipse.kura.message.KuraPayloadBuilder;
import org.eclipse.kura.data.DataService;
import org.osgi.service.component.annotations.Component;
import org.osgi.service.component.annotations.Reference;

@Component(service = KuraApplication.class)
```